

Lecture 7:

AI504 - Knowledge Representation

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1 Verbs

We live in a world with 2 types of creatures

- (a) moonacks
- (b) woodchucks

We know all moonacks are woodchucks, but nothing else

- (a) Do all who fear all moonacks fear all woodchucks?
- (b) Do all who fear all woodchucks fear all moonacks?
- (c) Do all who love all who fear all moonacks, love all who fear all woodchucks?
- (d) Do all who love all who fear all woodchucks, love all who fear all moonacks?

– in this example 1 and 3 are unknown, while 2 and 4 must be true.

Verbs are labels as

fears
→

- (a) Let $m = \{1\}$, $w = \{1, 2\}$, where $3 \xrightarrow{\text{fears}} m$

Then $\{1\} \subseteq \emptyset \perp$

- (b) Has to be true

- suppose i fear all woodchucks

Let m be an arbitrary moonack, m must be a woodchuck

Therefore I fear m

- Therefore i fear all moonacks

- (c) Let $m = \{1\}$, $w = \{1, 2\}$, where $3 \xrightarrow{\text{fears}} m$ and $4 \xrightarrow{\text{loves}} 3$

Then $\{1\} \subseteq \emptyset \perp$

- (d) Has to be true

- suppose i love an arbitrary person who fears all woodchucks

Let m be an arbitrary moonack, m must be a woodchuck

this arbitrary person must fear m

Since I love this person

- I must love all who fears all moonacks

This is confusing 3 should be true, and false should be false